

Name: _____

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Score: _____

Directions: This is a take-home test. It is due at the beginning of class on Monday, November 13. Please answer all questions in the space provided. Consider working the problems on scratch paper, then rewriting them neatly on the test. Additional copies of this test can be downloaded from my web page if needed.

For this test, you may discuss the problems among yourselves and share ideas, but the work you turn in must be your own (not copied). At the end of your solution to each problem, please list who (if anyone) you talked to about that problem, plus any additional information you want me to know (i.e. that you *gave* more help than you *received*, or vice versa, etc.).

- In order to get full credit, you must show all of your work.
- If you use a calculator, you must still write your solution step-by-step, as if it were done by hand.
- When appropriate, indicate your final answer clearly by putting it in a box.
- Constants that are not integers should be expressed as fractions.
- You may consult your text and notes, but **no** other source.
- Each problem is worth 10 points; if a problem has multiple parts, each part is 10 points.

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1. Suppose A is an $n \times n$ matrix and λ is an eigenvalue for A . Let $W = \{\mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} = \lambda\mathbf{x}\}$.
(In words, W consists of all eigenvectors corresponding to λ , including $\mathbf{0}$.)
Show that W is a subspace of $\mathbb{R}^n$.

2. Suppose A and B are $m \times n$ matrices. Is the set $W = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = B\mathbf{x}\}$ a subspace of $\mathbb{R}^n$?
Why or why not?

3. Decide if the set $\{(-4, 3, 3), (1, -2, 1), (1, 0, 1), (3, 2, 1)\}$ spans $\mathbb{R}^3$.

4. Is the set $\left\{ \begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 3 \\ 0 & 1 \end{bmatrix} \right\}$ a basis for $M_{2,2}$?

5. This problem concerns the matrix $A = \begin{bmatrix} -3 & 6 & -1 & 1 & 7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$.

(a) Find a basis for the row space of A .

(b) Find a basis for the column space of A .

(c) Find a basis for the nullspace of A .

6. Suppose a matrix A has 45 columns and its rows are linearly independent.

Also, the set $\{\mathbf{x} \in \mathbb{R}^{45} \mid A\mathbf{x} = \mathbf{0}\}$ is 7-dimensional. How many rows does A have? Explain.

7. For what values of x is the set $B = \{(1, x, x), (x, 1, x), (x, x, 1)\}$ **not** a basis for $\mathbb{R}^3$?

8. The set $B = \{1 - x + x^3, 1 + x, x + x^2, x^2 + x^3\}$ is a basis for P_3 . If $f = 5 + 8x + 3x^2 - 2x^3$, find $[f]_B$.